

	5.2×10^{-10}
6(a)	It is the amount of time required for a quantity of the radioactive isotope particles to decay to half its initial value.
6(b)	At the steeper part of the curve, the rate of decay of the particles is larger. Thus, it is more accurate to read the time taken for the number of particles to decay to half its original value at the steeper part of the graph.
6(c)	The total momentum of the products should be zero, since the nucleus was stationary initially. However, the momentum of the beta particle emitted has a vertical component, while that of the recoiled nucleus has no vertical components. This means that there must be another particle emitted such that its momentum has a vertical component that is equal in magnitude and opposite in direction to that of the beta particle.
6(d)(i)	${}^{90}_{38}\text{Sr} \rightarrow {}^{90}_{39}\text{Y} + {}^0_{-1}\text{e}$
6(d)(ii)	$A = \lambda N$ $\lambda = \frac{A}{N}$ $= \text{gradient of graph}$ $= \frac{4.6 \times 10^9 - 0}{6.0 \times 10^{18} - 0}$ $= 7.67 \times 10^{-10}$ $t_{\frac{1}{2}} = \frac{\ln 2}{\lambda} = 9.0 \times 10^8 \text{ s}$
6(e)	$N_J = N_{0,J} \times \left(\frac{1}{2}\right)^{3.5}$ $= 0.08839 N_{0,J}$ $N_K = N_{0,J} - N_J$ $= (1 - 0.08839) N_{0,J}$ $= 0.91161 N_{0,J}$ $\frac{N_J}{N_K} = \frac{0.08839}{0.91161} = 0.0970$
7(a)(i)	(0.22 – 0.32)